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Causal effects of coupon expiration on consumer acceptance: evidence from heterogeneous treatment effects in an observational scenario survey

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ABSTRACT

This paper examines how coupon expiration time affects consumer acceptance using observational scenario-survey data and causal machine learning. Because the mechanism assigning the expiration attribute is undocumented and responses are stated acceptance in hypothetical scenarios, we treat the two-hour (versus one-day) window as an observational exposure identified under selection-on-observables, supported by cross-fitted propensity scores, balance diagnostics, and sensitivity analysis. The Context-Contingent Time Preference (CCTP) framework supplies directional predictions about situational moderation. Inverse-probability weighting estimates the two-hour window reduces stated acceptance by 18.3% points (cluster-robust SE $= 0.014$); propensity score matching yields ATT $= -0.160$ (5,305 pairs). On a held-out confirmation sample, the penalty is pervasive but amplified by urgency ($\hat{\beta} = -0.087$, SE $= 0.041$) and destinations over 25 minutes away ($\hat{\theta} = -0.147$, SE $= 0.041$), most severe at the intersection of solitude and urgency (-0.134, SE $= 0.032$), and attenuated by social presence ($+0.088$, SE $= 0.022$; composition-adjusted $+0.047$, SE $= 0.032$). It concentrates in restaurant and coffee-house coupons and is absent for bar coupons. Simulations show effect-based targeting reduces conversion losses (29–42%) versus random allocation; propensity-based targeting performs worse than random. An *E*-value of 2.13 benchmarks sensitivity. Results concern stated, not revealed, acceptance.

KEYWORDS

Conditional Average Treatment Effect (CATE); causal machine learning; coupon expiration; heterogeneous treatment effects; stated preference; effect-based targeting

JEL CLASSIFICATION

M30; M31; M37; L17; L11

I. Introduction

Research background and motivation

Under the stable preference paradigm, coupon response is regarded as a fixed individual trait, formed by family characteristics and an internal reference price (Bawa and Shoemaker 1987; Gupta 1988; Mazumdar, Raj, and Sinha 2005). However, this static framework cannot explain why the same consumer might exhibit a negative response to a time constraint when travelling alone yet show a substantially milder response in a social environment: the treatment effect appears to differ in magnitude across circumstances, and understanding this variation is a prerequisite for designing time-limited promotions that do not destroy baseline demand.

Personalized promotion is now a cornerstone of marketing analytics, but its effectiveness is undermined by a fundamental asymmetry: targeting based on baseline purchase intention or 'propensity' classification often yields disappointing incremental effects, particularly when treatment effects are heterogeneous across service environments (Ascarza 2018). This propensity-effect divergence implies that predicting who is likely to buy reveals little about whose behaviour will change in response to specific promotional attributes, and that sensitivity to time pressure may be endogenous to situational factors rather than a fixed trait.

Research problem and questions

(1) How is the coupon expiration attribute (a two-hour versus one-day window) associated with stated acceptance decisions, and how does this association interact with situational factors?
(2) How does the severity of the temporal penalty vary across pre-specified consumer subgroups, and which situational contexts act as buffers against artificial urgency?
(3) Under binding budget constraints, how much incremental improvement can effect-

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based targeting bring over propensity-based and random allocation?

Because our data are observational scenario-survey responses, we frame all three questions in terms of conditional associations that admit a causal interpretation only under the assumptions made explicit in Section 3.3, and in terms of stated rather than revealed acceptance.
theoretical-framework-context-contingent-time-preference-cctp-as-an-organizing-framework.

Theoretical framework: Context-Contingent Time Preference (CCTP) as an organizing framework

To organize the heterogeneity analysis, we propose a Context-Contingent Time Preference (CCTP) framework. CCTP borrows the intuition of hyperbolic discounting – that immediate costs loom disproportionately large – and extends it by allowing the effective degree of time aversion to vary with scenario variables (social presence, destination urgency, travel distance). We emphasize at the outset that CCTP is used here as an organizing framework: it supplies directional, ex-ante predictions about which moderation contrasts should be positive or negative, but delivers no numerical predictions, and we do not interpret the estimated contrasts as structural parameters.

The framework proposes three psychological mechanisms, each generating a directional prediction. Social buffering reduces effective time aversion through distributed responsibility when travelling with peers (Rook and Fisher 1995), predicting a less negative effect in social contexts. Constraint accumulation amplifies temporal aversion when baseline urgency (e.g. commuting) coincides with an artificial deadline, predicting a more negative effect for urgency-bound trips. Spatial amplification increases temporal aversion when travel distance crosses the categories embedded in the study design, predicting a more negative effect for distant destinations. Together, these mechanisms imply that sensitivity to time pressure is context-dependent rather than fixed; our empirical objects are moderation contrasts on the probability scale, not discount-rate parameters.

Research contribution

Theoretical contribution

We formulate CCTP as an organizing framework linking situational configurations to directional predictions about treatment-effect heterogeneity, combining behavioural economics with normative evaluation theory (Rook and Fisher 1995). The framework addresses a limitation of static preference models – why the same coupon attribute carries different psychological costs in different consumption occasions – while stopping short of claiming that we recover structural time-preference parameters.

Methodological contribution

We demonstrate a disciplined workflow for causal machine learning on observational scenario-survey data: selection-on-observables identification with explicit assumptions, cluster-level resampling for inference in the absence of a respondent identifier, an exploration/confirmation sample split, pre-specified subgroup contrasts (H1, H2 and H4, with H3 explicitly exploratory) as the primary inferential objects, and transparent reporting of cross-estimator instability at the individual level.

Managerial contribution

We quantify, with corrected arithmetic, the value of effect-based targeting when the average effect of an intervention is negative: under binding budgets, CATE-based allocation reduces conversion losses relative to random allocation (point estimates of 29–42% loss reduction, with cluster-bootstrap intervals for the gap that are wide and, at some budgets, include zero) – materially smaller and more uncertain than previously claimed for this dataset – and we show that the policy problem is properly posed as minimum-harm allocation under an exactly binding budget, not as the choice of an 'optimal' budget.

II. Literature review

General research trend: from purchase prediction to causal inference

As personalized promotions have taken centre stage in modern marketing, the focus has evolved from predicting who will purchase to understanding the

causal effect of interventions. Discrete-choice models (McFadden 1974; Rossi, McCulloch, and Allenby 1996) capture correlation rather than causation when treatments are endogenously allocated, and strategies built on observational correlations can overstate causal effects through selection bias (Eckles and Bakshy 2021). Matching estimators imitate a randomized design by pairing units with similar covariate profiles (Rosenbaum and Rubin 1983), yet unobserved factors can bias estimates even after balancing (Heckman and Navarro-Lozano 2004); importantly, achieving balance on observed covariates does not itself validate the identifying assumption. Quasi-experimental methods such as Difference-in-Differences require parallel trends, which real-time contextual shocks readily violate (Card and Krueger 1994; Lechner 2011). Moreover, aggregate measures such as the Average Treatment Effect (ATE) can obscure segment-specific reactions in tailored marketing (Ascarza 2018).

For coupon expiration, traditional literature generally assumes a monotonic relationship: a shorter window increases perceived scarcity and urgency, accelerating purchase decisions (Aggarwal, Jun, and Huh 2011). These studies mainly rely on the 'cognitive load' paradigm, in which consumers under time pressure use heuristics to minimize cognitive effort (Wright 1974), and examine contexts where redemption requires minimal physical effort (e.g. one-click e-commerce flash sales). In contrast, the in-vehicle environment introduces a spatial-temporal 'double bind': the short expiration window is not just a digital countdown but is intrinsically linked to the logistical burden of physical movement, route deviation, and traffic mobility. When artificial urgency restricts autonomy in this physical manner, consumers may reject the promotion as an intrusion rather than an opportunity (Brehm 1966; Clee and Wicklund 1980). How situational factors such as spatial friction and social presence moderate responses to time constraints remains underexplored.

Machine learning for consumer behaviour analysis: from prediction to causal inference

Most machine-learning applications in and around consumer research are predictive rather than causal. Blanco-Oliver et al. (2024) use machine-learning methods to analyse the determinants of mobile-payment adoption, demonstrating the ability of flexible algorithms to rank behavioural drivers in complex adoption data. In an asset-pricing rather than consumer-behaviour setting, Wu, Zhou, and Su (2025) use machine learning on e-commerce review sentiment to predict stock-price movements, illustrating the predictive power of consumer-generated text for financial outcomes. These studies show that predictive ML does well in flexible approximation and variable importance ranking, but its reasoning goal is correlation and prediction, not the causal effect of operable attributes.

A smaller, more recent strand – Causal Machine Learning (CML) – targets treatment-effect heterogeneity directly. Causal Random Forests (CRF) (Wager and Athey 2018) adapt the random forest framework to estimate heterogeneous treatment effects without pre-specified interactions; meta-learners such as the T-learner flexibly model conditional response surfaces; and Double Machine Learning (DML) employs Neyman-orthogonal moments and cross-fitting to mitigate regularization bias (Chernozhukov et al. 2018). These methods estimate Conditional Average Treatment Effects (CATE) at the individual and subgroup levels, helping marketers identify which consumer groups respond most strongly to specific coupon attributes. For example, Chen et al. (2026) analysed a large-scale digital coupon project in China and found substantial response heterogeneity that traditional parametric models failed to capture.

Yet two cautions are warranted. First, CML is frequently deployed as a predictive segmentation tool: CRF and related methods identify heterogeneous subgroups but do not by themselves reveal the psychological mechanisms driving differential responses (Athey and Imbens 2019; Mullainathan and Spiess 2017). Second, individual-level CATE estimates are noisy predictions, and their rankings can be unstable across estimators and sample splits; subgroup-level contrasts and policy values are typically far more stable inferential targets. Our research solves these two problems: we use CCTP to specify a small set of directional adjustment contrasts in advance, use CML to estimate the retained data, and clearly report the reliability and unreliability of individual level inference.

Research gap and connection to methodology

Although previous studies have recognized that time preference shows substantial individual heterogeneity (Cohen et al. 2020; DellaVigna 2009), and that promotion effects vary by context (Chandon, Wansink, and Laurent 2000), these literatures remain theoretically disconnected. Existing models treat time aversion as a stable trait or random shock and cannot explain how real-time scenario configuration systematically changes the same individual's time valuation. Three specific gaps follow.

Gap 1: fragmented theorization and the missing configurational lens

Inman and McAlister (1994) showed that coupon expiration dates affect redemption behaviour, and subsequent work identified promotional context as a moderator; however, this literature never endogenized time aversion into situational configurations, leaving unexplained why the same consumer exhibits divergent reactions across occasions. Rook and Fisher (1995) documented the social facilitation of impulse buying without linking it to time discounting. Reactance theory (Brehm 1966; Clee and Wicklund 1980) explains constraint aversion but not how contextual buffers (peers, path flexibility) mitigate it. The CCTP framework organizes these fragments into a small set of directional, jointly testable predictions.

Gap 2: mechanism opacity in causal machine learning

Recent CML applications (Chen et al. 2026) successfully detect high-dimensional heterogeneity but deploy these algorithms as predictive black boxes (Athey and Imbens 2019; Mullainathan and Spiess 2017): without an external theoretical framework they cannot distinguish between opportunity-cost adaptation, reactance, or cognitive-load interpretations. We fill this gap by using CML to estimate theoretically interpretable moderation contrasts (α, β, θ) that link situational variables with predicted directions of response, while being explicit that these contrasts are descriptive decompositions of the CATE function, not structural parameters.

Gap 3: disciplined inference on heterogeneous effects under observational assignment

Applied work with scenario-survey data sometimes inherits the vocabulary of experimental design, such as 'randomized within scenarios', 'quasi-experimental validation' that lacks documentary support for the assignment mechanism. We take the opposite approach: we assume only selection on observables, we report the diagnostics that bear on its plausibility (covariate balance, weight distributions, sensitivity benchmarks), and we clearly separate exploratory heterogeneity discovery from confirmatory evaluation via a cluster-level sample split.

By combining behavioural theory with nonparametric estimation under stated assumptions, this study uses CML as a disciplined estimation engine for theory-guided contrasts. Four testable hypotheses follow, each expressed as a directional prediction about a moderation contrast on the stated-acceptance probability scale:

Hypothesis 1 (Social Buffering). *The negative effect of a short expiration window is attenuated when consumers travel with companions: the CATE is less negative in social contexts than in solitary contexts ($\alpha > 0$).*

Hypothesis 2 (Constraint Accumulation). *Baseline destination urgency amplifies the temporal penalty: among solitary travellers, the negative effect is stronger for work-bound trips than for flexible trips ($\beta < 0$).*

Hypothesis 3 (Detrimental Intersection). *The most severe resistance arises at the intersection of solitude and high baseline urgency: the CATE for solo travellers heading to work is substantially more negative than the CATE for socially accompanied travellers heading to no urgent destination (τ(Solo, Work) - τ(Social, NoUrgent) < 0).*

Hypothesis 4 (Spatial Amplification). *Travel distance amplifies the temporal penalty*

for destinations in the farthest travel-time category of the study design ($\theta < 0$).

III. Data and methodology

Data source, sample construction, and missing data

We use the In-Vehicle Coupon Recommendation dataset from the UCI Machine Learning Repository (dataset 603), which contains 12,684 scenario responses and 26 variables. Respondents were presented with hypothetical driving scenarios and asked whether they would accept a coupon shown under each scenario; the outcome Y_i is therefore **stated acceptance in a hypothetical scenario** (1 = stated acceptance, 0 = stated rejection), not observed redemption or purchase. The treatment attribute T_i indicates whether the coupon shown in the scenario carried a two-hour expiration (44.2%) versus a one-day expiration (55.8%). The raw baseline stated-acceptance rate is 56.8%.

Sample construction proceeds in three documented steps (Table 1). First, we exclude the car variable from the covariate set because it is missing for 99.15% of records; no imputation strategy is defensible at that missingness rate, and the variable is plausibly unrelated to the situational mechanisms of interest. Second, five behavioural-frequency variables (visit frequencies to bars, coffee houses, carry-out venues and two restaurant price tiers) carry between 0.84% and 1.71% missing values each (Supplementary Table S1); because these variables enter both the propensity-score and outcome models, we perform complete-case deletion, removing 605 rows and leaving 12,079 complete cases. Third, we remove 72 exact duplicate rows, yielding an **analysis sample of N = 12,007**. Keeping the 72 duplicates changes the cross-fitted IPW ATE from −0.183 to −0.190 (Section 4.1).

Respondent proxy and clustering

The dataset does not contain a responder identifier, although the basic survey yielded multiple scenario responses for each responder. As a conservative survey object, we cluster the observation results through accurate demographic data, repeating the unique combination of age, gender, marital status, parental identity, education, occupation and income, and get 490 data clusters. All inference (standard error, boot process, and the following example split) is performed at the cluster level. This is conservative relative to row-level inference but imperfect: a respondent may span multiple clusters, so standard errors may remain somewhat anti-conservative (Section 5.5).

Exploration/confirmation split

To separate the heterogeneity discovery from the confirmatory assessment, we split the analysis sample 50/50 by profile cluster (all observations sharing a profile fall in the same half) using a seeded random assignment. Section 4.6 reports the stability of all confirmatory comparisons in the five alternative random partitions. The discovery sample is used to train the causal forest, the T-learner and all nuisance models (propensity score and outcome regressions). The confirmation sample is used to evaluate the pre-specified subgroup contrasts of Hypotheses 1–4, the coupon-type contrasts and the policy values in the simulation. Average-effect estimators (IPW, PSM) are estimated on the full analysis sample with cluster-robust inference, as they involve no data-driven model selection.

The covariate set follows the CCTP structure – situational moderators, demographic controls, and behavioural history – and comprises 90 indicator

Table 1. Sample construction.

Step	Action	Rows removed	Rows remaining
Raw data	UCI In-Vehicle Coupon Recommendation (dataset 603)	—	12,684
Variable screening	Drop car (99.15% missing) from covariate set	0 rows	12,684
Complete cases	Delete rows with missing behavioral-frequency variables (0.84%–1.71% each)	605	12,079
De-duplication	Remove exact duplicate rows	72	**12,007**

Notes: The analysis sample contains 490 unique demographic-profile clusters (Section 3.1.1), which serve as the clustering unit for all inference. Outcome: stated acceptance (mean 0.568). Treatment share (two-hour coupon): 0.442.

and continuous terms when expanded for estimation (full definitions in Appendix A).

Descriptive statistics and baseline patterns

Treatment exposure varies systematically across observed scenario attributes: drivers travelling with friends receive two-hour coupons more often than solo drivers, and drivers heading to no urgent destination receive them more often than work commuters (Appendix B). These patterns rule out naive, unadjusted comparisons and motivate covariate adjustment; they do not, by themselves, validate or invalidate any identification assumption. Full variable definitions appear in Appendix A.

Causal framework and identification

Potential outcomes

We adopt the potential outcomes framework. For each observation i, $Y_i(1)$ and $Y_i(0)$ denote stated acceptance under the 2-hour and 1-day coupon, respectively, and the observed outcome is

$$Y_i = T_i \cdot Y_i(1) + (1 - T_i) \cdot Y_i(0), \qquad (1)$$

where $T_i \in \{0, 1\}$ denotes the expiration attribute shown in the scenario. We estimate the (ATE) – the population mean effect of the 2-hour versus 1-day window on stated acceptance – and Conditional Average Treatment Effects (CATE), which capture heterogeneity across contexts:

$$\begin{aligned} ATE &= E[Y_i(1) - Y_i(0)], \\ CATE &= E[Y_i(1) - Y_i(0) | X_i = x]. \end{aligned} \qquad (2)$$

Identification: an observational scenario survey

We state the data-generating process plainly: this is observational scenario-survey data. The original survey documentation does not record the mechanism by which the expiration attribute was assigned to scenarios – whether it was randomized within scenario blocks, rotated systematically, or otherwise determined. We therefore do not claim random or conditionally random assignment, and we identify causal effects under a selection-on-observables framework.

Assumption 1 (Conditional Independence, CIA). $\{Y_i(0), Y_i(1)\} \perp T_i \,|X_i$. The plausibility of CIA rests on the richness of the covariate set (situational, demographic and behavioural-history variables) together with the sensitivity analysis of Section 4.6; it cannot be established from the data. In particular, achieving covariate balance after matching or weighting is necessary for credible estimation but does **not** verify CIA, because balance diagnostics speak only to observed covariates.

Assumption 2 (SUTVA). $Y_i(T_i, T_{-i}) = Y_i(T_i)$ for all i – reasonable for independently presented hypothetical scenarios, though repeated responses by one person may share unobserved traits (addressed via cluster-level inference, Section 3.1.1).

Assumption 3 (Overlap). $0 < e(X_i) < 1$, where $e(X_i)$ is the propensity score. We verify overlap via the propensity-score distribution by treatment arm (Supplementary Figure S1) and assess post-adjustment covariate balance against the conventional |SMD| <0.1 reference threshold, reporting any exceedances rather than treating the threshold as a pass/fail gate: with the cross-fitted score, one of 90 covariates exceeds it, at 0.123, and all remaining covariates are at or below 0.072 (Appendix B).

Because the causal reading of every estimand in this paper is conditional on CIA, Section 4.6 quantifies how strong unmeasured confounding would need to be to explain away the headline estimate.

From theory to estimation: operationalizing CCTP as moderation contrasts

Classical exponential discounting (Samuelson 1937) posits a stable discount factor $\delta\tau$ with constant per-period discounting. Hyperbolic specifications instead use $\delta(\tau) = 1/(1 + \rho\tau)$, under which immediate costs weigh disproportionately. The CCTP framework adopts the hyperbolic form – we note that $1/(1 + \rho\tau)$ is a hyperbolic discount function rather than Samuelson's exponential formulation – and, more importantly, treats the effective aversion parameter as context-dependent rather than

fixed. We stress that this theory supplies directional predictions only: the signs of moderation contrasts, not their magnitudes, and we do not estimate or 'calibrate' the structural discount parameter ρ.

We operationalize CCTP through the following descriptive decomposition of the CATE function:

$$\tau(X) = \tau_0 + \alpha \cdot Social_i + \beta \cdot Urgency_i + \theta \cdot Distance_i + \eta(X_i), \quad (3)$$

where $Social_i$, $Urgency_i$, and $Distance_i$ are indicator variables for the corresponding situational states, $\eta(X_i)$ captures residual nonparametric interactions, and (α, β, θ) are moderation contrasts in probability-point units – differences between subgroup average treatment effects – not structural parameters of a discount-rate function. (A separate solitude coefficient would be perfectly collinear with the intercept, since $Solo_i = 1 - Social_i$). Equation (3) is an estimable linear projection of the CATE onto the theory-relevant moderators; its coefficients are the inferential targets of Hypotheses 1–4.

Test of H1 (social buffering)

We estimate subgroup average effects for $Social_i = 1$ (friends/partner/kids present) versus $Social_i = 0$ (alone) on the confirmation sample, pooling each subgroup average over the destination distribution within that social group (destination-share-weighted, with delta-method standard errors), testing whether the pooled social average exceeds the pooled solo average and quantifying the buffer as

$$\hat{\alpha} \approx \hat{\tau}_{Social} - \hat{\tau}_{Solo} \quad (4)$$

Because the destination mix differs mechanically across social contexts in the scenario design, we additionally report the composition-adjusted contrast conditional on the No Urgent Place destination, so that genuine moderation can be separated from destination composition (Section 4.2.1).

Test of H2 (constraint accumulation)

Operationalization in Appendix G.

Test of H3 (detrimental intersection)

We test the contrast $\hat{\tau}$(Solo, Work) - $\hat{\tau}$(Social, NoUrgent), predicted negative. Identification limits are stated plainly in Appendix G.

Test of H4 (spatial amplification as threshold heterogeneity)

The dataset records travel time in three ordinal categories (<15 min, 15–25 min, ≥ 25 min). We estimate the interaction model

$$Y_i = \beta_0 + \beta_1 T_i + \beta_2 \mathrm{Far}_i + \theta(T_i \times \mathrm{Far}_i) + \gamma' X_i + \varepsilon_i \quad (5)$$

where $\mathrm{Far}_i = 1$[travel time ≥ 25 min] and $\hat{\theta} = \hat{\tau}(\geq 25\,\text{min}) - \hat{\tau}(<25\,\text{min})$ is the additional penalty in the farthest category. We interpret $\hat{\theta}$ strictly as **distance-category heterogeneity, not a regression discontinuity**: travel time is recorded in coarse design categories, so there is no continuous running variable, no bandwidth to select and no density or continuity tests that could support an RDD design. The causal interpretation of $\hat{\theta}$ is conditional on the same CIA as all other estimands; it receives no additional 'local randomization' justification.

Estimation strategy

Average effects (benchmark)

We estimate the ATT via 1:1 nearest-neighbour propensity score matching **with replacement** and a caliper 0.2 SD of the propensity-score logits:

$$\widehat{ATT}_{PSM} = \frac{1}{N_1} \sum_{i:T_i=1} (Y_i - Y_{m(i)}), \quad (6)$$

We estimate the ATE via the IPW estimator,

$$\widehat{ATE}_{IPW} = \frac{1}{N} \sum_{i=1}^{N} \left(\frac{T_i Y_i}{e(X_i)} - \frac{(1 - T_i) Y_i}{1 - e(X_i)} \right), \quad (7)$$

with cluster-robust standard errors and balance/weight diagnostics in Appendix B. Estimation details are given in Appendix G.

Heterogeneous effects via causal machine learning

The primary CATE estimator is the Causal Random Forest (CRF) (Wager and Athey 2018),

which adapts recursive partitioning to conditional treatment effects:

$$\hat{\tau}_{CRF}(X_i) = \frac{1}{B}\sum_{b=1}^{B}\hat{\tau}_{l_b}(X_i), \tag{8}$$

As a second, architecturally distinct estimator, we use a T-learner metalearner (specification in Appendix G). Cross-estimator agreement is used diagnostically (Appendix C), not as a validity claim. All CATE models are trained on the discovery sample and evaluated on the confirmation sample. The division of labour between the confirmatory linear interaction models and the causal machine-learning estimators is summarized in Appendix G.

Scope of inference

Individual-level CATE estimates are noisy predictions whose rankings can vary across estimators and sample splits (Appendix C). We therefore restrict inference to (i) pre-specified subgroup contrasts motivated by CCTP (Hypotheses 1–4), (ii) coupon-type-specific average effects, and (iii) policy values.

Policy simulation design: minimum-harm allocation under a binding budget

The policy simulation evaluates the managerial value of moving from propensity-based targeting (predicting who is likely to accept) to effect-based targeting (predicting whose stated acceptance changes least under the two-hour window). Because the average effect of the two-hour coupon is negative throughout the population segments we estimate, the policy problem must be posed correctly: if all conditional effects are non-positive, the unconstrained optimum is the empty set – no one should receive the two-hour coupon. The operationally meaningful question is instead a minimum-harm allocation problem: given a managerial requirement to distribute two-hour coupons to exactly BN consumers (e.g. because a campaign has already committed to a short-window format), which allocation minimizes the conversion loss relative to the all-one-day baseline? We therefore do not define an 'optimal budget' B*; the budget B is treated as exogenously binding.

Let B $\in$ (0,1) denote the mandated budget as a fraction of the population. The policy chooses a set S with $|S| = BN$ to receive two-hour coupons (all others receive one-day). We compare the strategies in Table 2.

For any policy S, uplift is defined relative to the baseline policy in which everyone receives the one-day coupon and is estimated with doubly robust (DR) scores on the confirmation sample; the DR nuisance functions are logistic regressions, deliberately a different learner family from the gradient-boosting T-learner, and the random-allocation baseline is computed analytically (construction details in Appendix F):

$$\widehat{Uplift}(S) = \hat{A}(S) - \hat{A}(\emptyset). \tag{9}$$

where $\hat{A}(\emptyset)$ is the DR estimate of the all-one-day acceptance rate (0.634 in the confirmation sample), and the gap Uplift(S_{CATE}) – Uplift(S_{Random}) measures the incremental benefit of harm-minimizing allocation (Appendix F).

Robustness and sensitivity strategy

Our robustness framework (Section 4.6) comprises: matching/weighting diagnostics (balance, effective sample size, weight distributions); stability of subgroup conclusions under cluster-level resampling and alternative learner configurations, summarized qualitatively; cross-estimator agreement diagnostics; and an E-value sensitivity benchmark (VanderWeele and Ding 2017) for the confounding strength needed to explain away the average effect.

Table 3 (end of Section 4.5) summarizes the estimands, estimators, samples and inference modes used below.

Table 2. Description of allocation strategies.

Strategy	Selection rule	Description
Random	Uniform draw of BN	Budget-feasible benchmark ignoring all information
Propensity-based	Top-BN by $\hat{\mu}_0(X_i)$	Targets those with the highest estimated baseline acceptance probability
CATE-based	Top-BN by $\hat{\tau}(X_i)$	Targets those with the least harmful estimated effect

Table 3. Analysis map – estimators, samples and inference.

Estimand/analysis	Estimator	Sample	N	Inference
ATE	IPW (Equation 7)	Full analysis sample	12,007	Cluster-robust SE (490 clusters)
ATT	PSM 1:1 NN, caliper 0.2 SD, with replacement (Equation 6)	Matched sample	5,305 treated/5,305 matches (2,832 unique controls)	Cluster bootstrap (Abadie and Imbens 2008)
H1 contrast ($\hat{\alpha}$)	Covariate-adjusted linear interaction model (Equation 3), pooled and composition-adjusted	Confirmation half	Cluster-level random split	Cluster-robust SE
H2 contrasts ($\hat{\beta}$)	Same saturated model, urgency contrasts	Confirmation half	Cluster-level random split	Cluster-robust SE
H3 contrast (exploratory)	Same saturated model, two-cell contrast	Confirmation half	Cluster-level random split	Cluster-robust SE
H4 threshold contrasts ($\hat{\theta}$)	Three-category covariate-adjusted interaction (Equation 5)	Confirmation half	Cluster-level random split	Cluster-robust SE
Coupon-type effects	Stratified estimation	Confirmation half, per venue type	See Table E1 (Appendix E)	Cluster-robust SE
BLP projection	Projection of CRF CATE on moderators	Confirmation half	Cluster-level split	Cluster-robust SE
Policy values	Doubly robust scores	Confirmation half	Cluster-level split	Cluster bootstrap
E-value	VanderWeele and Ding (2017)	Point estimate of ATE	—	Sensitivity benchmark

Multiplicity

The confirmatory family consists of five pre-specified primary endpoints: the average effect (ATE) and the pre-specified hypothesis contrasts α, β_{Work}, β_{Home}, and θ. The H3 cell contrast is exploratory (Section 2.3) and is explicitly excluded from the confirmatory family; coupon-type effects, the BLP projection and policy-simulation values are likewise secondary or exploratory. We apply no formal family-wise adjustment; instead we limit the primary family to these five endpoints and report exact p-values throughout. With five primary endpoints, a Bonferroni threshold of $0.05/5 = 0.01$ would leave every primary conclusion reported below unchanged except the work-urgency contrast $\hat{\beta}_{\text{Work}}$ ($p = 0.032$), which would fall just outside the adjusted threshold – a further reason we describe H2's support as confined to the work destination and as the weakest of the significant results.

IV. Empirical results

All estimates below concern stated coupon acceptance in hypothetical scenarios, are conditional on the CIA of Section 3.3.2, and use cluster-level inference on the analysis sample ($N = 12{,}007$; 490 profile clusters) unless otherwise noted.

Average effects

Table 4 reports benchmark estimates of the average effect of the two-hour window on stated acceptance.

All four estimates in Table 4 point the same way: the 2-hour expiration window is associated with a large reduction in stated acceptance relative to the 1-day window, which is about 18.3% points at the population level (ATE) and 16.0 points among the treated (ATT). The cross-fitted specification prevents in-sample propensity

Table 4. Average effects and covariate balance.

Estimator	Estimand	Estimate	Inference	Sample/N
IPW, cross-fitted propensity score (primary)	ATE	−0.183	Cluster-robust SE = 0.014; 95% CI [−0.209, −0.156]; $p < 0.001$	Full analysis sample, $N = 12{,}007$
IPW, in-sample propensity score (robustness)	ATE	−0.188	Cluster-robust SE = 0.012	Full analysis sample, $N = 12{,}007$
IPW, keeping the 72 duplicate rows (robustness)	ATE	−0.190	Cluster-robust SE = 0.014	$N = 12{,}079$
PSM (1:1 NN, caliper 0.2 SD, with replacement; cross-fitted score)	ATT	−0.160	Cluster bootstrap (Abadie and Imbens 2008)	5,305 treated/5,305 matched controls (2,832 unique)

Notes: Balance and weighting diagnostics (Appendix B, Table B1; Supplementary Tables S2–S3): with the cross-fitted propensity score, the maximum absolute standardized mean difference across all 90 covariates is 0.123-one covariate (production occupations) marginally exceeds the conventional 0.1 threshold, and all remaining covariates are at or below 0.072; the in-sample variant achieves max |SMD| = 0.057. Effective sample sizes are 3,793 (treated) and 5,532 (control); the maximum IPW weight is 48.1 (a single observation) with a 99th percentile of 6.34, so weight concentration is limited to a thin tail. Matching retains all treated observations under the 0.2 SD caliper.

overfitting from compressing the weight distribution, yielding a slightly smaller point estimate (−0.183) than the in-sample variant (−0.188), and we adopt it as primary. The smaller absolute ATT relative to the ATE is consistent with the observed selection pattern (socially accompanied drivers), who exhibit milder penalties (Section 4.2.1), are more likely to receive 2-hour coupons (Appendix B), so the treated pool is disproportionately composed of consumers with built-in social buffers. This remains a conditional association: its causal reading rests on the untestable CIA, and Section 4.6 quantifies the confounding strength required to overturn it.

Heterogeneous treatment effects

Testing H1: social buffering ($\alpha > 0$)

Table 5 reports subgroup average effects by social context from the saturated covariate-adjusted interaction model of Equation (3), estimated on the confirmation sample with cluster-robust inference. Because H1 is a statement about social background, we report the average value of marginalized social background in the destination distribution of each social group, rather than a comparison based on a single destination unit.

H1 is **supported in its pooled form**: averaging over the destinations each group actually travels to, the two-hour penalty is 8.8% points milder under social presence than under solitude ($p < 0.001$). About half of the total gap is destination composition – socially accompanied trips in the design never head to work, the destination with the most severe penalty – and the remaining composition-adjusted moderation is +0.047: positive, as CCTP predicts, but imprecise at the confirmation-sample size ($p = 0.147$) – suggestive evidence consistent with normative evaluation theory (Rook and Fisher 1995), not an established moderation effect.

Testing H2: constraint accumulation ($\beta < 0$)

Table 6 reports destination-urgency contrasts for solitary travellers from the same model, isolating urgency from social confounding while adjusting for the full covariate set.

Two findings stand out. First, the temporal penalty does not require objective time pressure: solo travellers heading nowhere urgent already show a −14.8 pp penalty, so the 2-hour window is broadly aversive rather than binding only under time

Table 5. Social-buffering contrast (H1).

Quantity	Estimate	SE	*p*-value	CCTP directional prediction
Solo travellers, destination-pooled	−0.191	0.018	<0.001	High resistance
Social travellers, destination-pooled	−0.103	0.018	<0.001	Buffered
Pooled contrast $\hat{\alpha}$ (Social - Solo)	**+0.088**	0.022	<0.001	$\alpha > 0$: supported
Composition-adjusted contrast (conditional on No Urgent Place)	+0.047	0.032	0.147	directionally consistent, not statistically decisive
Destination-composition component of the pooled contrast	+0.041	—	—	social trips are never work commutes

Notes: Confirmation sample ($N = 6{,}024$; 245 clusters); single saturated covariate-adjusted linear interaction model with main effects for Social, Work and Home plus the full control set (the control set excludes the travel-time category indicators, which are modelled separately as the H4 moderators in Table 8); cluster-robust inference. Social denotes any accompanied state (friends/partner/kids); solo denotes travelling alone. Pooled averages are destination-share-weighted linear combinations of the model's cell estimates, with weights equal to the within-group destination shares and standard errors from the delta method applied to the same coefficient covariance matrix; all cell estimates in Tables 5 and 6 are linear combinations of the same coefficient vector, so the tables are mutually consistent by construction. The pooled contrast decomposes exactly as +0.088 = +0.047 (composition-adjusted moderation) + +0.041 (destination composition); the composition term arises because, in the scenario design, work commuting – the destination with the most severe penalty – is structurally a solo destination (42.9% of solo trips head to work; no socially accompanied trips do).

Table 6. Urgency contrasts among solo travellers (H2).

Destination	Subgroup CATE/contrast	SE	*p*-value	CCTP directional prediction
No Urgent Place (baseline cell)	$\hat{\tau}_{\text{NoUrgent}} = -0.148$	0.031	<0.001	Moderate penalty
Home	$\hat{\beta}_{\text{Home}} = -0.013$	0.033	0.697	$\beta < 0$: not supported
Work	$\hat{\beta}_{\text{Work}} = -0.087$	0.041	0.032	$\beta < 0$: supported
Implied $\hat{\tau}$(Solo, Work)	−0.235	0.029	<0.001	Most resistant solo cell

Notes: Same model and sample as Table 5. Contrasts are relative to the No Urgent Place cell among solo travellers; the implied Work-cell CATE is obtained as the linear combination $\hat{\tau} = \hat{\beta}_T + \hat{\beta}_{\text{TW}}$ of the same coefficient vector, with the delta-method standard error from the cluster-robust covariance matrix. The Work-cell standard error (0.029) is smaller than the reference-cell standard error (0.031) because the Work cell is considerably larger (n = 1,483 versus n = 603) and the two coefficients covary negatively; an independent check fitting the same specification on the solo subsample alone yields $\hat{\tau}$(Solo, Work) = −0.238 (SE = 0.030), corroborating the saturated-model estimate.

Table 7. Intersection contrast (H3) and scenario-design observability.

Panel A: Cell contrast (exploratory)			
Cell		CATE	SE
$\hat{\tau}$(Solo, Work)		−0.235	0.029
$\hat{\tau}$(Social, No Urgent)		−0.102	0.018
Contrast $\hat{\tau}$(Solo,Work) -$\hat{\tau}$ (Social,NoUrg)		−0.134	0.032 ($p < 0.001$)
Panel B: Passenger × destination observability in the scenario design			
Passenger	No Urgent Place	Home	Work
Alone	observed	observed	observed
Partner	observed	observed	**not observed in scenario design**
Friend(s)	observed	**not observed in scenario design**	**not observed in scenario design**
Kid(s)	observed	observed	**not observed in scenario design**

Notes: Confirmation sample; cluster-robust inference; all Panel A quantities from the same saturated model as Table 5 and 6 (−0.235 - (−0.102) = −0.133 ≈ −0.134 up to rounding, so the tables reconcile exactly). The Social × Work cell is structurally empty (zero observations) in the scenario design; consequently the Passenger × Destination × Treatment three-way interaction is not identified, and H3 is evaluated as the direct two-cell contrast in Panel A. We do not impute or extrapolate to unobserved cells.

Table 8. Distance-category heterogeneity in the temporal penalty (H4).

Quantity	Estimate	SE	*p*-value	Interpretation
$\hat{\tau}_{near}$ (<15 min)	−0.147	0.020	<0.001	Penalty for the nearest category
$\hat{\tau}_{mid}$ (15–25 min)	−0.104	0.018	<0.001	Penalty for the intermediate category
$\hat{\tau}_{far}$ (≥ 25 min)	−0.294	0.038	<0.001	Penalty for the farthest category
$\hat{\theta}_{mid} = \hat{\tau}_{mid} - \hat{\tau}_{near}$	+0.043	0.023	0.067	Intermediate-category difference (n.s.)
$\hat{\theta}_{far} = \hat{\tau}_{far} - \hat{\tau}_{near}$	−0.147	0.041	<0.001	Additional penalty in the ≥ 25-min category

Notes: Confirmation sample ($N = 6{,}024$); covariate-adjusted interaction model with main effects for the distance categories and the full control set; cluster-robust inference. As an aggregation consistency check, the share-weighted average of the three distance-cell CATEs (44.8%/43.5%/11.7% near/mid/far) is −0.145, versus the full-sample IPW ATE of −0.183; the gap reflects the different estimand, sample and conditioning sets, and we read the stratified estimates as effect distributions across cells, not as decompositions of the ATE.

constraints. Second, baseline urgency amplifies the penalty specifically when the destination is work: the work-bound contrast (−8.7 pp, $p = 0.032$) is significant, while home-bound trips are indistinguishable from flexible trips (−1.3 pp, n.s.). Therefore, H2 is supported in the urgency of its binding type of work, that is, constraining the accumulated CCTP tags, but it is not suitable for milder destinations.

Testing H3: the detrimental intersection

Table 7 reports the H3 contrast and the observability structure of the full passenger × destination matrix. H3 is an exploratory hypothesis (Section 2.3); we report it with the same inferential machinery but flag its post-hoc status.

The most severe stated resistance occurs where CCTP predicts: solo, work-bound travellers (−0.235), while socially accompanied flexible travellers show a significantly milder penalty (−0.102). The 13.4-point gap between the two poles of the gradient (SE = 0.032) supports the intersection logic of H3, subject to its exploratory status. We emphasize what is not claimed: because the Social × Work cell does not exist in the design, we cannot decompose this gap into separable main effects and interactions, and we make no claim about how socially accompanied work commuters would have responded.

Cross-estimator agreement and the scope of inference

Agreement between the Causal Forest and the T-learner at the individual level is moderate, and well above chance: the Spearman rank correlation is 0.471, the top-decile overlap 41% (against 10% under independence), and the top-30% overlap 74%. Detailed agreement statistics, the learner-family policy-value check, and the statement of how these agreement levels discipline the scope of inference are reported in Appendix C, and the Best Linear Predictor projection is in Table C1.

Table 9. Policy simulation results by binding budget (uplift versus all-one-day baseline).

Budget B	Random (analytic)	Propensity-based	CATE-based	CATE versus random: gap [95% CI]
20%	−0.0277 [−0.0347, −0.0204]	−0.0339 [−0.0491, −0.0213] (sel. CATE −0.1695)	−0.0160 [−0.0293, −0.0003] (sel. CATE −0.0800)	+0.0117 [+0.0002, +0.0243]; 42% loss reduction
30%	−0.0416 [−0.0520, −0.0306]	−0.0524 [−0.0699, −0.0357] (sel. CATE −0.1746)	−0.0296 [−0.0464, −0.0115] (sel. CATE −0.0988)	+0.0120 [−0.0038, +0.0262]; 29% loss reduction
50%	−0.0693 [−0.0866, −0.0510]	−0.0761 [−0.0969, −0.0535] (sel. CATE −0.1522)	−0.0495 [−0.0715, −0.0258] (sel. CATE −0.0989)	+0.0199 [+0.0021, +0.0362]; 29% loss reduction

Notes: DR policy values on the confirmation sample (N = 6,024; 245 clusters); baseline $\hat{A}(\emptyset) = 0.634$. DR nuisance functions are logistic regressions (a different learner family from the gradient-boosting T-learner used in the cross-estimator comparison), with propensity scores trimmed to [0.05, 0.99]. The random baseline is analytic (B × mean DR uplift), so it carries no simulation noise. Brackets are 95% cluster-level bootstrap intervals (500 replications, resampling demographic-profile clusters); gap intervals come from the same bootstrap, and gap point estimates are the displayed CATE-minus-random differences computed from unrounded values (they may differ from the difference of the rounded entries by ± 0.0001). CATE rankings average three causal-forest seeds. 'sel. CATE' is the average DR-estimated individual uplift among selected consumers; the identity Uplift(S) = B × sel. CATE holds exactly for DR scores. Cross-estimator consistency check: the DR estimate of the ATE on the confirmation sample is −0.139 (so B × ATE(DR) at B = 20% is −0.028, the analytic random baseline), versus the full-sample IPW ATE of −0.183. The 4.4pp difference between these two average-effect estimates decomposes into +3.4pp from the subsample (the IPW ATE restricted to the confirmation half is −0.148) and +1.0pp from the estimator (DR versus IPW on the same confirmation half); we report both sources rather than describe the two scales as interchangeable.

Threshold heterogeneity analysis (H4)

Table 8 reports the three-category distance analysis of Equation (5) on the confirmation sample, with the intermediate category now estimated explicitly.

The two-hour penalty for destinations at least 25 minutes away (−29.4 pp) is roughly twice the penalty in the nearest category (−14.7 pp), and the category contrast $\hat{\theta}_{\text{far}} = -0.147$ is precisely estimated ($p < 0.001$). As stated in Section 3.4, we interpret this result strictly as distance-category heterogeneity, not a regression discontinuity.

One descriptive feature sharpens the interpretation: the gradient is non-monotone across the first two categories: the intermediate category (15–25 minutes) is slightly *less* penalized than the nearest one ($\hat{\theta}_{\text{mid}} = +0.043$, n.s.), so the penalty does not grow smoothly with travel time; the large change occurs as a category jump at the design's 25-minute boundary. We return to this pattern when assessing competing explanations (Appendix D). heterogeneity-by-coupon-type

Heterogeneity by coupon type

Effects vary systematically by coupon venue type: the temporal penalty concentrates in sit-down and planned-consumption venues (full-service restaurants −19.3 pp and −17.0 pp, carry-out −14.7 pp, coffee houses −13.8 pp), while bar coupons show no detectable penalty (−0.6 pp, n.s.) – the only venue type where the two-hour window does not reduce stated acceptance. Stratified estimates and their interpretation are reported in Appendix E (Table E1).

Policy simulation results

Table 9 reports doubly robust policy values on the confirmation sample, with uplift measured relative to the all-one-day baseline ($\hat{A}(\emptyset) = 0.634$). All values are negative because the two-hour window reduces stated acceptance for every segment we can estimate; the exercise is the minimum-harm allocation problem of Section 3.6.

Three conclusions follow. First, CATE-based allocation attains the least harm at every budget, with point estimates of **29–42% loss reduction** relative to random allocation, a material but imprecisely estimated gain. The bootstrap intervals for the CATE-versus-random gap are wide – at B = 20% the lower bound (+0.0002) barely excludes zero, and at B = 30% the interval includes zero. We therefore describe the targeting gain as directionally consistent across budgets but only moderately decisive statistically; this uncertainty is itself a substantive finding, since it bounds how much any targeting rule can rescue from a negatively acting attribute. Second, propensity-based targeting performs worse than random allocation at every budget (e.g. −0.052 versus −0.042 at B = 30%): baseline acceptance is not merely uninformative about treatment responsiveness but positively misleading. This is an especially sharp instance of the

propensity–effect divergence documented by Ascarza (2018): high-baseline users typically operate in low-urgency situations, where the penalty is moderate, so they are not the least-harmed segment under the two-hour window. Third, even the best allocation at B = 20% still forfeits 1.6pp of stated acceptance relative to the all-one-day baseline; targeting mitigates, but cannot eliminate, the harm of a negatively acting attribute.

Two implications follow. First, there is no interior 'optimal budget' B*: because all effects are negative, any expansion of the mandated budget adds harm, and the flat region simply means that harm accumulates slowly up to roughly half the population. Second, no fixed budget cap (such as B* = 30%) can be defended on these estimates; the defensible managerial statement is that the low-marginal-harm region extends to approximately 50–60% of the population, beyond which each additional two-hour coupon becomes more damaging.

Robustness and sensitivity

Matching and weighting

Balance and weight diagnostics are reported in Table 4 and Appendix B; we report both the cross-fitted and in-sample propensity-score variants rather than choose silently. **Sample-split stability**. Every confirmatory contrast was re-estimated on five alternative cluster-level splits; all signs and directional conclusions are unchanged, and the primary-split H3 estimate is in absolute value, the smallest of the six splits, so the intersection contrast is, if anything, conservative on the primary split (ranges in Appendix B, Table B1).

Sensitivity to unobserved confounding (E-value)

Translated to the risk-ratio scale, the IPW estimate corresponds to RR = 0.718 – computed as $(p_0 + \text{ATE})/p_0$ using the IPW-weighted untreated mean ($p_0 = 0.647$) and the IPW ATE (−0.183), both derived from the same cross-fitted propensity scores on the full analysis sample, yielding an **E-value of 2.13** (VanderWeele and Ding 2017). An unmeasured confounder associated with both receipt of the two-hour coupon and stated acceptance by a risk ratio of 2.13-fold each, above and beyond the measured covariates, could fully explain away the estimate, while weaker confounding could not. We present this as a **sensitivity benchmark, not as proof of robustness**: confounders of that strength are not implausible in principle (e.g. situational mood or brand attachment correlated with scenario attributes), and the benchmark applies to the average effect rather than to each subgroup contrast. Discussion

V. Discussion

Mechanisms of heterogeneity: the importance of context

The population-level association is substantially negative (ATE = −0.183), yet this average hides substantial situational variation that stable individual characteristics cannot easily explain, but that the CCTP anticipates in direction.

Constraint accumulation amplifies resistance

Work-bound urgency adds a significant −8.7 pp on top of the already-substantial solo penalty ($\hat{\beta}_{\text{Work}}$, p = 0.032), producing the most resistant cell (−0.235), while home-bound trips add nothing detectable (−1.3 pp, n.s.). In the CCTP reading, what amplifies temporal aversion is obligation-backed urgency colliding with an artificial deadline (Brehm 1966), not destination directedness per se; the work/home asymmetry – both are directed destinations, but only commuting carries a binding schedule – is the sharpest single piece of evidence for the constraint-accumulation mechanism over a generic directedness account (Mazumdar, Raj, and Sinha 2005).

Competing explanations

We confront three alternative accounts of the heterogeneity patterns, marking which generate ex-ante directional predictions and which are ex-post interpretations. The three accounts are assessed in Appendix D: the scarcity-heuristic account is directionally falsified, the linear opportunity-cost account is not supported in its linear form, and selection on unobservables is reduced but cannot be eliminated.

Table 10. Marginal CATE by budget.

Budget B	10%	30%	50%	60%	80%
Marginal CATE	−0.157	−0.179	−0.198	−0.208	−0.232

Notes: Confirmation sample; consumers ranked by CRF-estimated CATE averaged over three seeds (least harmful first). The marginal curve is computed from causal-forest predictions, whose average on the confirmation sample (−0.201) is more negative than the DR-based uplift scale used in Table 9; the two objects answer different questions (forest-predicted individual effects versus DR policy values), so exact numerical correspondence between the tables is not expected – the qualitative flat-then-steep shape is the shared content. Supplementary Figure S3 plots the full marginal curve across all budgets.

Theoretical integration and scholarly contextualization

The integration of these findings with the stable-preference paradigm and with the machine-learning literature is developed in Appendix D.

Consequences for management

The dark side of urgency

The negative average effect cautions against uniform two-hour deployments: short windows reduce stated acceptance across nearly every context we can estimate, with no segment showing a positive estimated effect at the subgroup level. Uniform urgency tactics cannibalize baseline acceptance rather than activating engagement.

Precision targeting as risk mitigation, not growth

Under binding budgets, harm-minimizing allocation by estimated CATE reduces conversion losses relative to random allocation (point estimates of 29–42%, with wide cluster-bootstrap intervals; Table 9), and the low-marginal-harm region extends to about 50–60% of the population before marginal damage rises steeply (Table 10). Managerially: (i) if a short-window campaign is mandated, prioritize socially accompanied, flexible, near-destination consumers and bar/carry-out occasions; (ii) avoid two-hour coupons for solo work-bound travellers and distant destinations, where penalties reach −24 to −29 pp; (iii) treat the budget as a constraint to be minimized against, not an 'optimal' scale to be chosen – the all-one-day policy remains first-best in these data.

Limitations and future directions

Stated preferences

The outcome is stated acceptance in hypothetical scenarios, which need not coincide with revealed redemption or purchase; hypothetical bias could attenuate or inflate the estimated penalties, and all magnitudes should be read as stated-preference effects.

Identification

The assignment mechanism for the expiration attribute is undocumented; identification rests on CIA, supported by a rich covariate set, balance diagnostics and a sensitivity benchmark, none of which is dispositive. Observable imbalance after adjustment does not validate CIA, and the E-value is a benchmark rather than a guarantee.

Cluster proxy

Without a respondent identifier, we cluster on 490 demographic-profile duplicates. This is conservative but imperfect; if respondents span multiple clusters, standard errors may remain somewhat anti-conservative.

Hypothesis status. H3 is an exploratory, data-informed contrast and does not inherit the pre-specification discipline that protects H1, H2 and H4; the discovery/confirmation split cannot retroactively convert a data-informed hypothesis into a pre-registered one. The H3 contrast should therefore be read as a well-powered exploratory analysis that future work should test with truly pre-registered designs.

Design-imposed limits

The coarse travel-time categories preclude any discontinuity design and limit functional-form conclusions about distance; the structurally empty Social × Work cell prevents decomposition of the intersection contrast; and the binary expiration contrast (2 h versus 1 d) cannot trace the dose–response curve of window length.

Future research

Four directions follow. (1) Through the field experiment of the actual transaction, it is verified whether the preference gradient, especially the large individual elastic penalty and coupon gap, can survive in the incentive compatibility measurement. (2) Continuous expiration windows to locate the window length at which urgency transitions from neutral to detrimental. (3) Distributional analysis: uniform short-window campaigns may burden low-income solo shoppers facing stricter time budgets disproportionately; income-stratified CATE estimation in the present data was insufficiently powered at the cluster level, and these potentially regressive consequences merit dedicated study. (4) Sequential allocation designs in which expiration policies adapt to response history and inventory levels connecting the static contrasts estimated here to dynamic promotion policy.

VI. Conclusion

This study re-examines how coupon expiration time relates to consumer acceptance using observational scenario-survey data and causal machine learning, with stated acceptance in hypothetical scenarios as the outcome and selection-on-observables as the identification framework. Three main conclusions can be drawn.

First, the 'urgency premium' finds no support in these data: the two-hour window reduces stated acceptance by 18.3% points on average relative to the one-day window (IPW ATE = −0.183, 95% CI [−0.209, −0.156]), with a matched ATT of −0.160. The two estimators converge on a large, pervasive penalty for artificial short deadlines in physically effortful redemption contexts.

Second, the penalty is pervasive but systematically amplified by context. Social presence offsets close to half of the pooled solo penalty (pooled $\hat{\alpha} = +0.088$, $p < 0.001$), with roughly half of that gap attributable to destination composition and the composition-adjusted remainder (+0.047) imprecisely estimated; obligation-backed urgency amplifies resistance – solo work commuters exhibit the most severe penalty (−0.235), 13.4 points below socially accompanied flexible travellers, while home-bound trips add nothing detectable; distant destinations ($\geq$ 25 minutes) roughly double the penalty ($\hat{\theta}_{\text{far}} = -0.147$), a pattern we interpret as distance-category heterogeneity rather than a regression discontinuity; and the penalty is concentrated in restaurant and coffee-house coupons while bar coupons show no detectable effect.

Third, effect-based targeting helps, but modestly and with considerable uncertainty. Simulations show CATE-based allocation reduces conversion losses relative to random allocation under binding budgets (point estimates of 29–42%; the cluster-bootstrap interval includes zero at the intermediate budget (B = 30%)), with marginal harm flat through roughly half the population and steep thereafter; because all estimated effects are negative, the policy problem is minimum-harm allocation under a mandated budget, not the choice of an 'optimal' budget, and the all-one-day policy remains first-best.

Author contributions

CRediT: **Xingtian Si:** Formal analysis, Methodology, Software, Validation, Writing – original draft, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

The data that support the findings of this study are openly available from the UCI Machine Learning Repository at https://archive.ics.uci.edu/dataset/603/in+vehicle+coupon+recommendation (dataset 603). All analysis code – sample construction, estimation, sample splitting, policy simulation, and sensitivity analysis – is provided as supplementary material to enable full reproduction of every reported number.

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Appendices

Appendix A. Variable definitions

CCTP-aligned covariate structure. Covariates are organized into three categories: (i) situational moderators: passenger (Alone Friend(s) Kid(s) Partner), destination urgency (No Urgent Place Home Work), time of day, weather, temperature, direction match and travel-time category; (ii) demographic controls: age, gender, income, education, occupation, marital status and parenthood; and (iii) behavioural history: visit frequencies to bars, coffee houses, carry-out venues and restaurants. The full covariate set expanded for estimation comprises 90 indicator and continuous terms.

The outcome Y_i is stated acceptance (0/1). Treatment T_i indicates a two-hour (versus one-day) expiration. Situational variables: time of day {7 AM, 10 AM, 2 PM, 6 PM, 10 PM}; destination {No Urgent Place, Home, Work}; passenger {Alone, Friend(s), Kid(s), Partner}; weather {Sunny, Rainy, Snowy}; temperature {30, 55, 80 °F}; travel time to the coupon venue, recorded by the scenario design in three ordinal categories (<15 minutes; 15–25 minutes; $\geq$ 25 minutes), encoded by the two nested indicators toCoupon_GEQ15min and toCoupon_GEQ25min (the raw file also contains toCoupon_GEQ5min, which equals 1 for every record and is excluded as degenerate); direction match (same/opposite). Demographics: gender, age (8 brackets), marital status (5 categories), education (6 categories), occupation (25 categories), income (9 brackets), parenthood. Behavioural history: visit frequencies to bars, coffee houses, carry-out venues and restaurants (<$20; $20–$50), each in {never, less than 1, 1–3, 4–8, more than 8 per month}; the car variable is excluded (99.15% missing). Expanded with indicators, the estimation covariate set comprises 90 terms.

Appendix B. Selection patterns, propensity score and balance diagnostics

(A) **1. Observed selection into the two-hour treatment**. Treatment prevalence differs by scenario attributes: socially accompanied drivers (especially those with friends) receive two-hour coupons more often than solo drivers, and flexible-destination drivers more often than work commuters. These differences motivate covariate adjustment; they are properties of the scenario-assignment process, whose mechanism is undocumented (Section 3.3.2).

(B) **2. Balance and weighting diagnostics**. The full 90-covariate balance table appears in Supplementary Table S2, the matching/weighting summary in Supplementary Table S3, and the covariate-by-covariate Love plot in Supplementary Figure S2.

Weight-tail sensitivity. Because cross-fitting widens the weight distribution relative to the in-sample variant, we also checked whether the average effect hinges on the thin upper weight tail: trimming the propensity score to [0.05, 0.95] gives ATE = −0.185 (maximum weight 20.0), winsorizing the weights at their 99th percentile gives −0.195, and dropping the single maximum-weight observation gives −0.187, all of them within about one percentage point of the primary −0.183.

Estimator and specification robustness. The ATE is −0.183 with cross-fitted propensity scores, −0.188 with in-sample scores, and −0.190 when the 72 duplicate rows are retained. Subgroup conclusions are unchanged under cluster-level resampling and under alternative learner configurations (alternative tree counts, leaf sizes and honesty fractions for the CRF; alternative base learners for the T-learner), which we summarize qualitatively rather than tabulate.

Unobserved-confounding benchmark: descriptive reference. As a descriptive reference point, the strongest observed covariates exhibit risk-ratio associations of at most about 1.2–1.3 with *both* treatment assignment and stated acceptance (unadjusted margins; Supplementary Table S5). These are not the conditional benchmarks formally required by the E value, but they indicate the order of magnitude that unmeasured confounding factors need to exceed. The appropriate conclusion is moderate – the Title Association will require non trivial, but it is conceivable that the unmeasured confusion is false and the CIA remains a hypothesis, not a discovery.

Balance and weight diagnostics in detail. With the cross-fitted propensity score, the post-IPW maximum |SMD| is 0.123 across 90 covariates (one covariate marginally above 0.1; all others $\leq$ 0.072). IPW effective sample sizes are 3,793 (treated) and 5,532 (control). The maximum weight is 48.1 for a single observation (99th percentile 6.34). The in-sample propensity variant is tighter (max |SMD| = 0.057; max weight 10.7) at the cost of potential overfitting bias in the weights.

Table B1. Covariate balance and weight diagnostics (cross-fitted propensity score).

Diagnostic	Value
Covariates in balance check	90
Max |SMD| after IPW (cross-fitted PS)	0.123 (1 covariate > 0.1; all others $\leq$ 0.072)
Max |SMD| after IPW (in-sample PS, robustness)	0.057
IPW effective sample size, treated	3,793
IPW effective sample size, control	5,532
Max IPW weight	48.1 (single observation)
99th-percentile IPW weight	6.34
PSM matched pairs (1:1 NN, caliper 0.2 SD, with replacement)	5,305/5,305
Unique controls used in matching	2,832

Notes: Balance on observed covariates is a necessary estimation diagnostic; it does not validate the conditional independence assumption, which concerns unobservables.

Table B2. Confirmatory contrasts across the primary split and five alternative cluster-level splits.

Contrast	Primary split	Range across five alternative splits
$\hat{\alpha}$ (H1, pooled social buffer)	+0.088	[+0.02, +0.12]
$\hat{\beta}_{\text{Work}}$ (H2)	−0.087	[−0.04, −0.17]
$\hat{\theta}_{\text{far}}$ (H4)	−0.147	[−0.13, −0.20]
H3 contrast (exploratory)	−0.134	[−0.13, −0.18]

Notes: Each alternative-split range brackets the corresponding primary-split estimate. For the H3 contrast the primary-split estimate (−0.134*) sits at the lower edge of the alternative-split range and is, in absolute value, the smallest of the six splits we computed, so the intersection contrast is if anything conservative on the primary split. The imprecision of the composition-adjusted α (H1) is likewise stable across splits.*

Appendix C. Cross-estimator agreement and Best Linear Predictor (BLP) projection

Cross-estimator agreement in detail. Agreement between the Causal Forest and the T-learner at the individual level is moderate and well above chance: the Spearman rank correlation is 0.471, the top-decile overlap 41% (against 10% under independence) and the top-30% overlap 74%. The two architectures thus agree on broad patterns while retaining residual disagreement about *which specific individuals* are most or least affected, which is expected for noisy individual-level predictions at this sample size. Crucially for a targeting exercise, however, the **policy values** induced by the two architectures are nearly identical: at B = 20%, 30% and 50%, allocations ranked by the T-learner attain doubly robust policy values within 0.2–0.4% points of the CRF-ranked allocations (Supplementary Table S4). The DR nuisance functions here are logistic regressions, a different learner family from the gradient-boosting T-learner, so the evaluation is not mechanically favourable to it; the targeting conclusions therefore do not depend on the choice of learner even where individual rankings differ. This residual individual-level instability disciplines our inference: all confirmatory conclusions in this paper rest on subgroup contrasts (Sections 4.2.1–4.2.3; pre-specified for H1, H2 and H4, exploratory for H3) and on policy values (Section 4.5), which are stable under cluster-level resampling, rather than on individual-level rankings. We accordingly do not report the distribution of individual CATE estimates or threshold-crossing shares, as such quantities are not reliably estimable at the individual level in these data.

We do not interpret the shape of the individual CATE distribution, and we do not report shares of individuals above or below effect thresholds, since such shares are artefacts of prediction noise rather than estimable population quantities.

Table C1. BLP projection of individual CATE estimates on CCTP moderators (confirmation sample; descriptive).

Moderator	Coefficient
Social	+0.028
Work	−0.030
Far ($\geq$ 25 min)	−0.010

Notes: Projection of the multi-seed CRF CATE estimates onto the theory-relevant moderators. Because the dependent variable is itself a prediction (a generated regressor), conventional standard errors are not valid for inference here – we verified that cluster-robust SEs are an order of magnitude smaller than the seed-to-seed variability of the underlying CATE estimates (Section 3.6) – so we report the projection as a **descriptive sign-and-magnitude summary only**. The signs align with the subgroup contrasts of Sections 4.2–4.3 (social accompaniment less negative; work urgency more negative; far distance more negative), which remain the primary inferential evidence for H1–H4.

Appendix D. Competing explanations and theoretical integration

(a) **Scarcity heuristics and cognitive load (ex-ante prediction, directionally falsified)**. Scarcity/urgency literature predicts that a shorter window increases acceptance by triggering competition cues and heuristic processing (Aggarwal, Jun, and Huh 2011; Wright 1974); under this account, the group facing the most objective time pressure – solo work commuters – should accept more readily, because the heuristic value of the deadline is highest for them. We observe the opposite: the penalty is most severe exactly in that cell (−0.235). The directional failure of this ex-ante prediction strengthens the case that, in physically effortful redemption contexts, artificial urgency is processed as a constraint on autonomy (reactance) rather than as a scarcity cue.

(b) **Linear time opportunity cost (ex-ante prediction, not supported in its linear form)**. A simple opportunity cost account predicts that the penalty should increase monotonically and smoothly with travel time, because a longer detour consumes more time. The category pattern is inconsistent with this: the medium-distance category (15–25 minutes) does not lie monotonically between the nearer and farther categories, and the larger change in the penalty occurs at the designed 25-minute category boundary rather than along a smooth gradient. Distance appears to interact with the temporal window as a feasibility constraint on the redemption plan, not as a metered time cost, though with three coarse categories we cannot characterize the true functional form.

(c) **Selection on unobservables (partially mitigated, not eliminated)**. Observed selection into the two-hour treatment is documented and adjusted (balance: max $|SMD| = 0.123$ post-IPW), and the E-value benchmark (2.13) showed that only confounding stronger than the observed covariates we measure, with a combined risk ratio of at least 2.13 with both treatment and outcome, could fully explain the average association. This partially mitigates selection concerns but cannot eliminate them: instantaneous states (hunger, mood, promotion fatigue) and brand attachment were not recorded. The competing-explanation status of selection is therefore 'reduced but open'.

We flag that the reactance and feasibility-constraint readings offered in (a) and (b) are ex-post interpretations of patterns we did not pre-register; they are consistent with CCTP's directional predictions but are not independent tests of them.

Theoretical integration. This research qualifies the 'stable preference paradigm' (Bawa and Shoemaker 1987; Gupta 1988) with situational evidence: stated coupon responsiveness changes systematically with transient context variables, undermining propensity-based targeting and supporting Ascarza (2018)'s propensity–effect divergence thesis. The mechanism evidence is consistent with this account: frequent, high-baseline consumers typically operate in low-urgency situations, where the artificial deadline is least destructive, so baseline propensity and treatment sensitivity diverge. By endogenizing time aversion into situational configurations, the emphasis shifts from 'who discounts?' to 'when do they discount?'. Methodologically, relative to predictive ML applications adjacent to consumer research (Blanco-Oliver et al. 2024; Wu, Zhou, and Su 2025) and to black-box deployments of causal ML (Chen et al. 2026), our contribution is a disciplined division of labour: theory supplies directional, pre-specified contrasts (for H1, H2 and H4; H3 is exploratory); causal ML estimates them flexibly; and inference is confined to the estimands (subgroup contrasts and policy values) that survive cluster-level resampling and cross-estimator comparison.

Appendix E. Heterogeneity by coupon type

The scenarios span five coupon venue types, and the composition of expiration windows differs across them: 55.7% of coffee-house coupons carry the two-hour window, versus only 22.9% of bar coupons. A pooled estimate would mix venue-specific effects with venue-specific assignment, so we stratify by coupon type (Table E1).

The temporal penalty is concentrated in sit-down and planned-consumption venues: full-service restaurants show the largest penalties (−19.3 pp and −17.0 pp), followed by carry-out (−14.7 pp) and coffee houses (−13.8 pp). **Bar coupons show no detectable penalty (−0.6pp, n.s.)** – the only venue type where the two-hour window does not reduce stated acceptance. A plausible ex-post reading is that bar visits in these scenarios are impromptu social consumption occasions for which a short window does not bind: the consumer is often already at or near the venue, and the decision to stop is made on the spot, so a two-hour deadline imposes no real constraint. By contrast, restaurant and coffee-house redemptions typically require deliberate detours and seating time, against which a two-hour window binds tightly. This venue-type gradient is, to our knowledge, a new stylized fact for this dataset and qualifies any uniform recommendation about short-window coupons.

Venue type as consumption-occasion heterogeneity. The coupon type gradient (Table E1) adds a dimension of demand: when redemption requires planned, time-consuming consumption (sit-down restaurants and coffee houses), the penalty is largest; when consumption is impromptu and immediate (bars), the penalty is smallest and statistically undetectable. This pattern is consistent with the benefit-congruency account of promotion effectiveness (Chandon, Wansink, and Laurent 2000): when redemption logistics are least congruent with the ongoing activity, time-limited coupons are most destructive.

Table E1. Coupon-type-specific effects of the two-hour window on stated acceptance.

Coupon type	Estimated effect	SE	Significance	Sample share
Coffee House	−0.138	0.024	p < 0.001	31.7%
Bar	−0.006	0.040	n.s.	15.9%
Carry out	−0.147	0.037	p < 0.001	18.6%
Restaurant (<$20)	−0.170	0.024	p < 0.001	22.0%
Restaurant ($20–$50)	−0.193	0.042	p < 0.001	11.8%

Notes: Confirmation sample; covariate-adjusted per-venue models with the full control set (excluding the travel-time category indicators, which are modelled separately as the H4 moderators in Table 8); cluster-robust inference. Two-hour treatment shares differ across venue types (e.g. Bar 22.9% versus Coffee House 55.7%), motivating stratified estimation. The share-weighted average of the venue-specific effects is −0.132, versus the full-sample ATE of −0.183; the difference reflects the estimand, sample, and conditioning-set gaps documented in Section 4.3, and we report it explicitly rather than treat the stratified estimates as an ATE decomposition.

Appendix F. Policy simulation details

With the candidate allocation strategies defined, we next make the evaluation criterion precise. For any policy S, the predicted overall stated-acceptance rate is estimated with doubly robust (DR) scores on the confirmation sample, and uplift is defined relative to the baseline policy in which everyone receives the one-day coupon. The DR nuisance functions are deliberately drawn from a **different learner family than the cross-estimator comparison**: logistic regressions for both outcome models and the treatment model (with scores trimmed to [0.05, 0.99]) versus the gradient-boosting T-learner – so the policy evaluation does not mechanically favour the learner whose rankings are being assessed:
where $\hat{A}(\emptyset)$ is the DR estimate of the all-one-day acceptance rate (0.634 in the confirmation sample). The random-allocation baseline is computed **analytically** as E[Uplift (Random at B)] = B × (mean DR uplift), which by construction equals B times the DR estimate of the ATE on the confirmation sample; this removes simulation noise from the benchmark entirely. A cross-sample consistency check against B × ATE_{IPW} is reported in Section 4.5. For each strategy we also report the **average estimated CATE among the selected consumers**, which lets the reader verify the identity Uplift(S) = B × (average CATE over S) directly from the table. The gap Uplift(S_{CATE}) – Uplift(S_{Random}) measures the incremental benefit of harm-minimizing allocation. Two stabilization measures address the run-to-run variability of forest-based CATE rankings (Appendix C): we average individual CATE estimates over three causal-forest seeds (median seed-to-seed pointwise SD = 0.008) before ranking, and we quantify the sampling uncertainty of every policy value with 500 cluster-level bootstrap replications of the confirmation sample, resampling demographic-profile clusters rather than rows.

Appendix G. Estimation and operationalization details

Propensity-score matching (ATT), where N_1 is the number of treated observations and m(i) denotes the matched control for treated observation i. Because matching is with replacement and observations are clustered, standard errors are computed by **cluster bootstrap** over profile clusters (Abadie and Imbens 2008).

IPW estimator (ATE) and propensity-score specification. The propensity score is an unpenalized logistic regression on the 90-term covariate set, **cross-fitted with five cluster-grouped folds** so that in-sample overfitting cannot compress the weight distribution; fitted scores are trimmed to [0.01, 0.99]. The in-sample (non-cross-fitted) variant is reported as a robustness check in Section 4.1.

Causal Random Forest. B is the number of trees and $\hat{\tau}_b$ is the within-leaf treatment-effect estimate of tree b.

As a second, architecturally distinct estimator, we use a T-learner metalearner, which fits separate flexible outcome models $\hat{\mu}_1(x)$ and $\hat{\mu}_0(x)$ and forms $\hat{\tau}_{\text{TL}}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x)$.

Test of H2 (Constraint Accumulation). Restricting to solitary travellers to isolate urgency from social confounding, we contrast destination categories: $\hat{\beta}_{\text{Work}} = \hat{\tau}(\text{Work}) - \hat{\tau}(\text{NoUrgent})$ and $\hat{\beta}_{\text{Home}} = \hat{\tau}(\text{Home}) - \hat{\tau}(\text{NoUrgent})$, predicted negative.

Test of H3 (Detrimental Intersection): identification limits. Two identification limits are stated plainly. First, the Social × Work cell is **structurally empty in the scenario design** (zero observations), so the full factorial cannot be completed and unobserved cells are marked accordingly (Table 7, Panel B). Second, the three-way interaction Passenger × Destination × (Social buffering × Constraint accumulation) is not identified; H3 is tested as a direct contrast between the two populated cells that theory designates as the poles of the resistance gradient.

Division of labour: the pre-specified subgroup contrasts of Hypotheses 1–4 are estimated with covariate-adjusted linear interaction models (Equations 3 and 5) on the confirmation sample, whose inference properties are transparent under cluster-robust estimation; the causal forest and T-learner supply individual-level CATE estimates used for cross-estimator diagnostics (Appendix C), the BLP projection (Appendix C), and the policy rankings in the simulation (Section 4.5).